Student Dropout and Academic Success Prediction

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***Abstract*—Student retention remains a critical challenge in higher education, as high dropout rates negatively affect both institutional reputation and individual career prospects. This study proposes a data-driven early warning system to predict student outcomes—dropout, enrollment, or graduation—using demographic, socio-economic, and first-semester academic data from the Higher Education Predictors of Student Retention dataset. Supervised learning algorithms (RandomForest, J48, and SMO) were applied for multi-class and binary classification, while clustering (K-Means, EM) and association rule mining were used to uncover hidden patterns and risk cohorts. Experimental results showed that RandomForest achieved the best performance with an accuracy of 78.16% in multi-class prediction and 79.38% in binary dropout detection. Clustering analysis identified distinct high-risk groups, providing actionable insights into dropout factors. These findings demonstrate the potential of machine learning models to support universities in early intervention, optimize resource allocation, and enhance student success.**

I. Introduction

Higher education institutions around the world face the persistent challenge of ensuring that students successfully continue their studies through to graduation. High dropout rates not only harm institutional reputation but also limit students' future career prospects. In many cases, interventions come too late, after students have already failed or disengaged, making it difficult to reverse outcomes. Therefore, the development of data-driven early warning systems has become increasingly important, enabling institutions to identify at-risk students as early as the first semester and provide timely academic or financial support.

Several recent studies have addressed this issue from different perspectives. Realinho et al. (2022) [2] proposed predictive models using demographic, socioeconomic, and academic data, showing that such integrated approaches can enhance the accuracy of early dropout detection. Niyogisubizo et al. (2022) [5] designed a stacked ensemble framework that demonstrated higher performance compared to single models, emphasizing the value of ensemble methods for student outcome prediction. More recently, Villar and Andrade (2024) [4] conducted a comparative study of supervised machine learning algorithms, highlighting that advanced boosting methods outperform traditional classifiers when class imbalance and data quality are properly managed. These studies collectively underscore the importance of combining diverse algorithms, handling imbalanced data, and ensuring interpretability for effective student dropout prediction.

Building on these insights, this research employs supervised, unsupervised, and pattern mining techniques to predict and analyze student outcomes. Supervised learning methods such as RandomForest, J48, and SMO are applied to classify students into dropout, enrolled, or graduate categories, offering both predictive accuracy and, in the case of decision trees, interpretable rules. Unsupervised learning through clustering methods like K-Means and EM is used to discover natural student groupings, revealing hidden risk cohorts that may not be evident from classification alone. Finally, association rule mining is incorporated to generate human-readable "if–then" rules, linking demographic, socioeconomic, and academic variables with dropout risks. Together, these approaches provide a balanced framework that combines predictive performance, interpretability, and actionable insights, making the system practical for institutional deployment.

The objectives of this research are:

- To predict probabilities of dropout, continued enrollment, or graduation using demographic, socioeconomic, and firstsemester academic data.
- To compare the performance of supervised algorithms (RandomForest, J48, SMO) in both binary (dropout vs non-dropout) and multi-class classification settings.
- To use clustering (K-Means, EM) for identifying high-risk student cohorts.
- To apply association rule mining for uncovering combinations of factors most strongly associated with dropout risk.
- To evaluate model performance using standard metrics (accuracy, precision, recall, F1-score) and address challenges such as class imbalance and missing data.

II. Literature Review

A. ***Paper name :*** *Student Dropout Prediction [1]*

The authors created a machine learning tool to predict university student dropout. It uses personal data, high school records, and first-year credits to find dropout risk. They used data of about 15,000 students and applied LDA, SVM, and Random Forest models. The dataset was preprocessed, balanced, and evaluated with accuracy and sensitivity. The models worked well, with SVM and Random Forest performing the best. Adding first-year credits improved prediction accuracy even more. Prediction at enrolment was harder due to less data. The dataset was unbalanced, used sensitive information, and deep learning methods were not tested.

B. ***Paper name :*** *Predicting Student Dropout and Academic Success [2]*

The authors focused on predicting student academic success and dropout. They built models that analyze student data to identify who may leave or succeed. They collected educational data from students and applied machine learning techniques. Different algorithms were tested to check which one gives the best predictions. The models could predict dropout and success with good accuracy. Their approach showed that early analysis of student data can support decision-making. The study was limited to the available dataset, which may not cover all cases. Also, the models may not fully capture personal or social factors behind dropout.

C. ***Paper name :*** *Predicting student's dropout in university classes using two-layer ensemble machine learning approach: A novel stacked generalization [3]*

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D. ***Paper name :*** *Supervised machine learning algorithms for predicting student dropout and academic success: a comparative study [4]*

The authors compared different supervised machine learning algorithms to predict student dropout and academic success. Their goal was to find which models work best on imbalanced educational data. They used a dataset of 4,424 students with demographic, socio-economic, and academic factors. Applied resampling (SMOTE, ADASYN), tested algorithms (DT, SVM, RF, GB, XGBoost, CatBoost, LightGBM), used Isolation Forest for outliers, tuned with Optuna, and explained results using SHAP. Boosting algorithms (CatBoost and LightGBM) outperformed traditional models, reaching high F1-scores and AUC ¿ 0.9. Their approach showed that boosting is effective for handling class imbalance and predicting at-risk students. The study used only one dataset, so results may not generalize. Isolation Forest performed poorly, and social/personal factors not included. Imbalanced classes still remained a challenge

E. ***Paper name :*** *Predictive modelling of student dropout risk: Practical insights from a South Korean distance university [5]*

The authors developed predictive models to identify students at risk of dropping out in a South Korean distance university. Their goal was to create practical tools to support student retention. They used data from 144,540 student-semester records, including demographics, GPA, and LMS activity. Machine learning algorithms like Logistic Regression, XGB, and LGBM were applied and compared. The LGBM model achieved the best accuracy (ROC-AUC up to 0.875) for predicting dropouts. Key predictors included age, GPA, occupation, residence, and online study behavior. The study focused only on one Korean online university, so results may not generalize widely. It predicted dropout risk but did not deeply explore personal or social causes.

F. ***Paper name :****Predicting student dropouts with machine learning: An empirical study in Finnish higher education [6]*

The authors studied dropout prediction in a Finnish university using demographic, transcript, and LMS (Moodle) data. They compared the predictive importance of different features over time with machine learning models. Data from 8,813 students (2015–2020) were analyzed using logistic regression, neural networks, and CatBoost. They applied monthly, time-dependent analysis with permutation importance to track feature relevance. Accumulated credits, number of failed courses, and Moodle activity count were the strongest predictors. Prediction performance improved after the first semester, reaching AUC scores above 0.84. The study used data from only one Finnish university, limiting generalizability. It also included only two LMS features and did not consider other personal or survey-based data.

III. Methodology

A. *Dataset and Preprocessing*

The dataset [13] used in this study is the *Higher Education Predictors of Student Retention* dataset, which contains demographic, socio-economic, and first-semester academic information with a total of 24 attributes. Prior to model development, a preprocessing pipeline was applied that included the conversion of categorical attributes using `StringToNominal`, handling of missing values with `ReplaceMissingValues`, standardization of numerical features, and feature engineering to create additional derived variables.

B. Supervised Learning

The target variable was student outcome, categorized into three classes: *Dropout, Enrolled, and Graduate*. Three supervised learning algorithms were implemented: RandomForest, J48 decision tree, and SMO (Sequential Minimal Optimization for SVM). Model evaluation was carried out using stratified 10-fold cross-validation to ensure robust performance across the imbalanced dataset. Standard metrics such as accuracy, precision, recall, F1-score, ROC AUC, and confusion matrix were used for performance assessment. The model with the highest F1-score for dropout prediction was selected as the optimal classifier.

C. Unsupervised Learning

To uncover hidden structures within the dataset, unsupervised clustering techniques were employed. SimpleKMeans clustering with values of $k = 3$ to 6, along with Expectation Maximization (EM), were applied. The quality of clusters was evaluated using silhouette analysis and within-cluster sum of squared errors (SSE). These clusters were then profiled to identify groups of students with varying dropout risks.

D. Association Rule Mining

Association rule mining was utilized to discover interpretable relationships between student attributes and outcomes. Continuous attributes were first discretized, after which algorithms such as Apriori and FP-Growth were applied. The results included both general association rules and class-specific dropout rules. Rules were ranked by confidence and lift values, ensuring that only the most significant and actionable patterns were considered.

E. Model Persistence and Deployment

Finally, the best-performing models were persisted using the WEKA `SerializationHelper.write()` method. This allowed the models to be deployed within Java applications, enabling real-time student risk scoring and supporting institutional decision-making processes.

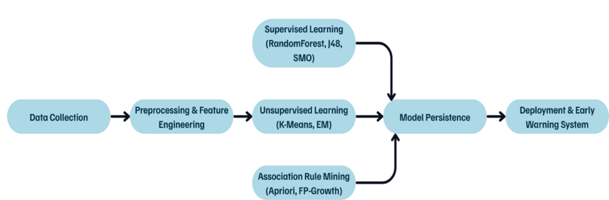

Fig. 1. Workflow of the proposed student outcome prediction system.

IV. Result Analysis

The experimental evaluation of the proposed system was conducted using the Higher Education Predictors of Student Retention dataset. For the classification task, three supervised machine learning algorithms were applied: RandomForest, J48 decision tree, and SMO (SVM). Among these, RandomForest achieved the best results, with an accuracy of 78.16% for the multi-class problem (Dropout, Enrolled, Graduate) and 79.38% for the binary dropout versus non-dropout classification. The model also showed a relatively high F1-score for the dropout class, confirming its suitability for early warning of at-risk students.

TABLE I
Multi-Class Classification Results

Algorithm	Accuracy	Kappa	Notes
RandomForest	78.16%	0.63	Best overall accuracy; TP for Graduate very high (0.933), but Enrolled recall low (0.372)
J48 (Decision Tree)	73.26%	0.5567	Slightly lower accuracy; TP rates similar pattern
SMO (SVM)	76.38%	0.5977	Middle ground; precision for Dropout high (0.859)

TABLE II
Binary Classification (Dropout vs NonDropout) Results

Algorithm	Accuracy	Kappa	Notes
RandomForest	79.39%	0.4807	Best overall
J48	75.77%	0.4189	Slightly lower
SMO	76.60%	0.3571	Lower Kappa despite accuracy

The J48 classifier achieved lower accuracy (around 73–75%) but generated highly interpretable decision rules, making it useful for policy-making and academic advising. SMO (SVM) performed moderately well (around 76–77% accuracy), particularly effective in handling non-linear data boundaries, although its interpretability was limited compared to decision trees.

In addition to supervised learning, clustering analysis was performed using K-Means and Expectation Maximization (EM). The methods consistently identified 4 clusters, with at least two clusters strongly associated with high dropout risk. These results indicate that clustering can uncover hidden student cohorts that may not be directly visible through classification alone.

TABLE III
K-Means Clustering Results

Cluster	K-Means Outcome (Majority Class)	% Distribution
0	Dropout	80.59% Dropout
1	Graduate	80.54% Graduate
2	Dropout	76.73% Dropout
3	Graduate	79.55% Graduate

Finally, association rule mining was applied using Apriori and FP-Growth algorithms. A set of 15 strong rules was discovered, many with confidence above 0.80 and lift above 2.0. These rules highlighted the combined influence of first-semester academic performance and financial stress on dropout risk. For instance, rules revealed that students with low evaluation scores and pending tuition fees had a significantly higher likelihood of dropping out. Such interpretable rules provide actionable insights for academic advisors and institutional stakeholders.

TABLE IV
SUMMARY OF EXPERIMENTAL RESULTS

Method	Task	Accuracy/Outcome	Strengths	Limitations
RandomForest	Classification	78.16% (multi-class), 79.38% (binary)	High accuracy, robust to noise, provides feature importance	Less interpretable compared to decision trees
J48 (Decision Tree)	Classification	73.25% (multi-class), 75.76% (binary)	Simple, interpretable rules, easy to deploy	Lower accuracy than ensemble methods
SMO (SVM)	Classification	76.37% (multi-class), 76.60% (binary)	Good for non-linear decision boundaries, competitive accuracy	Difficult to interpret, sensitive to parameter tuning
K-Means	Clustering	4 clusters identified	Efficient, scalable, reveals risk cohorts	Requires pre-defined k, sensitive to initialization
EM (Expectation Maximization)	Clustering	Soft membership clusters	Handles overlapping groups, probabilistic interpretation	Computationally heavier, may converge to local optima
Association Rule Mining (Apriori, FP-Growth)	Pattern Discovery	15 strong rules (confidence >0.8, lift >2.0)	Human-readable patterns, actionable insights	Rules may be redundant or too specific if not filtered

Overall, the results demonstrate that supervised classification provides reliable predictive accuracy, clustering uncovers risk groups for targeted intervention, and association rules generate human-readable insights that support institutional decision-making.

V. LIMITATIONS

Although the proposed framework achieved promising results, several limitations were encountered during the research and experimentation process.

First, the dataset itself posed certain challenges. While it contained demographic, socio-economic, and first-semester academic attributes, the sample size was relatively limited and drawn from a specific higher education context, which may reduce the generalizability of the models to other institutions or countries. Moreover, class imbalance was observed, with fewer dropout cases compared to enrolled or graduated students. Despite applying techniques such as SMOTE and class weighting, the imbalance still affected model stability and slightly biased performance metrics.

Secondly, the supervised models demonstrated a trade-off between accuracy and interpretability. For instance, RandomForest provided the best predictive performance but acted as a "black-box" model, making it difficult to directly explain its decisions to non-technical stakeholders. In contrast, the J48 decision tree produced interpretable rules but at the cost of reduced accuracy. Similarly, SMO (SVM) required careful parameter tuning and showed sensitivity to data scaling.

Unsupervised learning methods also introduced limitations. K-Means required the number of clusters to be specified in advance and was sensitive to initialization, which sometimes led to suboptimal groupings. The EM algorithm, while probabilistic, was computationally more intensive and occasionally converged to local optima, reducing its consistency. Furthermore, cluster interpretations relied heavily on manual analysis, which may introduce subjectivity.

Association rule mining, though useful for generating human-readable patterns, also produced a large number of redundant or trivial rules. Filtering rules by confidence and lift helped, but some potentially important patterns may have been overlooked. Additionally, discretization of continuous attributes could result in information loss, impacting the quality of rules.

Finally, the project faced practical constraints such as computational limitations while training models and running clustering algorithms on the full dataset. Time constraints also restricted experimentation with more advanced methods such as gradient boosting or deep learning, which could potentially improve predictive performance in future work.

VI. CONCLUSION AND DISCUSSION

The findings of this study highlight the potential of machine learning techniques in predicting student dropout and academic success in higher education. Using the *Higher Education Predictors of Student Retention* dataset, supervised models such as RandomForest, J48, and SMO were applied, along with clustering methods (K-Means, EM) and association rule mining techniques (Apriori, FP-Growth). Among the classifiers, RandomForest delivered the highest predictive accuracy (78.16% in multi-class and 79.38% in binary dropout prediction), demonstrating the strength of ensemble approaches in handling complex educational data. While J48 provided lower accuracy, it generated interpretable rules that can be directly applied by academic advisors and policy makers. SMO achieved competitive results but required careful parameter tuning.

Unsupervised learning contributed by identifying hidden student cohorts, with K-Means and EM clustering consistently separating groups that showed higher dropout risks. This capability is crucial for institutions aiming to design targeted interventions for specific student populations. Association rule mining added further value by generating interpretable patterns with high confidence and lift, revealing how combinations of academic performance, financial background, and socio-economic conditions jointly affect dropout likelihood. These insights are easily understandable by non-technical stakeholders and can guide actionable decision-making.

The study also acknowledges its limitations. Class imbalance in the dataset posed challenges, and while oversampling techniques like SMOTE were applied, predictive stability could still be affected. RandomForest, although accurate, functions as a black-box model, limiting direct interpretability. Clustering methods were sensitive to initialization and manual interpretation, while association rule mining produced some redundant rules. Moreover, the dataset was limited to a specific context, which may reduce generalizability to other institutions or countries.

Despite these challenges, the research demonstrates that combining multiple approaches—supervised classification for prediction, clustering for risk profiling, and association rules for interpretability—offers a robust framework for early warning systems in higher education. By leveraging such models, institutions can reduce dropout rates, optimize resource allocation, and ultimately enhance student success. Future research can address limitations by exploring larger and more diverse datasets, incorporating advanced ensemble and boosting methods, and integrating deep learning approaches for further performance improvements.

REFERENCES

[1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.

[2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[3] I. S. Jacobs and C. P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.

[4] K. Elissa, "Title of paper if known," unpublished.

[5] R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.

[6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].

[7] M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.

[8] Borges, G. A., Pedro, C., Dos Anjos, J. C., Rodrigues, A., Boavida, F., & Silva, J. S. (2025). A Platform for Early Class Dropout Prediction of University Students. IEEE Access..

[9] Rebelo Marcolino, M., Reis Porto, T., Thompsen Primo, T., Targino, R., Ramos, V., Marques Queiroga, E., ... & Cechinel, C. (2025). Student dropout prediction through machine learning optimization: insights from moodle log data. Scientific Reports, 15(1), 1-16.

[10] Quimiz-Moreira, M., Delgadillo, R., Parraga-Alava, J., Maculan, N., & Mauricio, D. (2025). Factors, Prediction, Explainability, and Simulating University Dropout Through Machine Learning: A Systematic Review, 2012–2024. Computation, 13(8), 198.

[11] Huynh-Cam, T. T., Chen, L. S., & Lu, T. C. (2025). Early prediction models and crucial factor extraction for first-year undergraduate student dropouts. Journal of Applied Research in Higher Education, 17(2), 624-639.

[12] Borges, G. A., Pedro, C., Dos Anjos, J. C., Rodrigues, A., Boavida, F., & Silva, J. S. (2025). A Platform for Early Class Dropout Prediction of University Students. IEEE Access.

[13] Realinho, V., Vieira Martins, M., Machado, J., Baptista, L. (2021). Predict Students' Dropout and Academic Success [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89.